

SMART CAMPUS EVENT MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering
By**

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May, 2026

BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled “**Smart Campus Event Management System**” by BORA BALARAJU (VTU25631) has been carried out in the Winter Semester of Academic Year 2025–2026 (WS2526).

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. I also declare that i have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

The Smart Campus Event Management System is a web-based application designed to automate and simplify the management of campus events, workshops, and seminars in educational institutions. Traditional event management methods are often manual, time-consuming, and prone to errors in handling registrations, updates, and communication.

This system provides a centralized platform where students can securely register, log in, browse upcoming events, register for events, and provide feedback. Administrators can create, update, delete, and monitor events through a dedicated dashboard.

The application includes features such as event scheduling, registration deadline management, department-based filtering, announcements, and email notifications. Students receive event details and registration confirmations directly to their registered email IDs.

Overall, the Smart Campus Event Management System offers a scalable, secure, and efficient digital solution for managing campus events while reducing manual effort and improving productivity.

Keywords: Smart Campus, Event Management, Spring Boot, Thymeleaf, MySQL, Automation.

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LIST OF ACRONYMS AND ABBREVIATIONS

CI	Continuous Integration
CD	Continuous Deployment
ORM	Object Relational Mapping
MVC	Model View Controller
API	Application Programming Interface
REST	Representational State Transfer
JPA	Java Persistence API
CRUD	Create, Read, Update, Delete
SMTP	Simple Mail Transfer Protocol
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language

CSS	Cascading Style Sheets
SQL	Structured Query Language
DBMS	Database Management System
JWT	JSON Web Token
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment
JSON	JavaScript Object Notation
SRP	Single Responsibility Principle
DTO	Data Transfer Object

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Chapter 1

INTRODUCTION

1.1 Introduction

The Smart Campus Event Management System is a web-based application developed to simplify and automate the management of campus events, workshops, and seminars in educational institutions. Traditional event management processes are often handled manually, which can lead to inefficiencies, communication delays, and errors in registration handling.

This system provides a centralized platform where students can securely register, log in, browse upcoming events, register for events, and provide feedback. Administrators can efficiently manage the complete event lifecycle by creating, updating, deleting, and monitoring events through a dedicated admin dashboard.

The system also includes advanced features such as event registration tracking, announcements, email notifications, department-based filter-

ing, analytics, and registration deadline management. By integrating these functionalities into a single platform, the application improves efficiency, reduces manual effort, and ensures better communication between students and administrators.

Overall, the Smart Campus Event Management System offers a scalable, secure, and professional solution for managing campus events in a digital environment.

1.2 Aim of the Project

The aim of this project is to develop a web-based Smart Campus Event Management System to automate the management of campus events, workshops, and seminars. It provides a centralized platform for students to register and participate in events, while admins can efficiently create, update, and manage events. The system reduces manual work, improves communication, and ensures better event handling.

1.3 Scope of the Project

The scope of this project includes managing campus events through a centralized web platform. Students can register, view, and participate in events, while admins can manage event creation, updates, and registrations. The system can be further enhanced with mobile support, cloud deployment, and advanced analytics.

1.4 Framework

The Smart Campus Event Management System is developed using Spring Boot with a layered architecture following the Model-View-Controller (MVC) design pattern. The frontend is designed using HTML5, CSS3, and Thymeleaf, while the backend uses Spring MVC and Spring Data JPA for business logic and database operations. MySQL is used for storing event, user, and registration details, ensuring efficient data management.

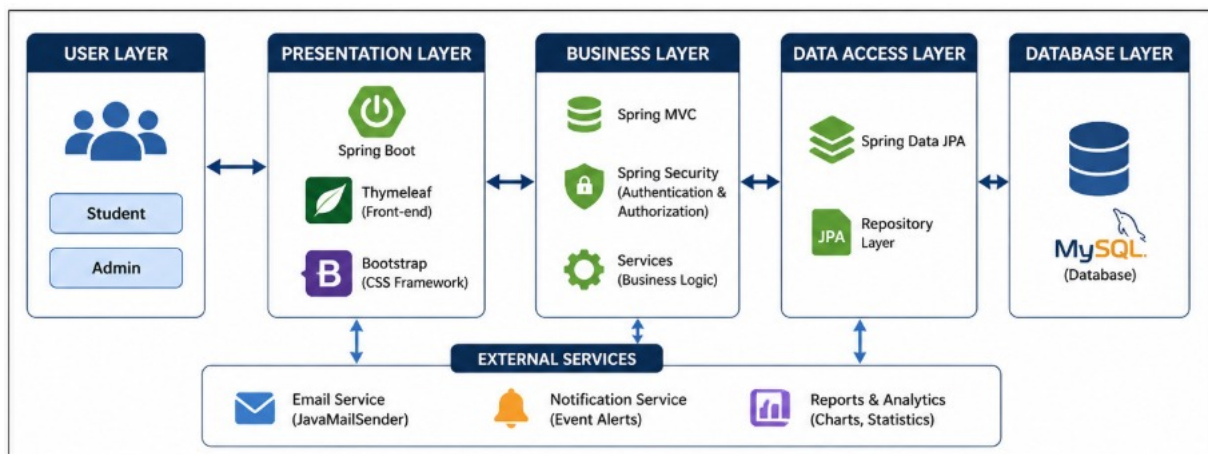


Figure 1.1: Framework Architecture

Figure 1.1 illustrates the framework architecture of the Smart Campus Event Management System. It shows the interaction between frontend, backend, and database layers. The architecture ensures modularity, scalability, and smooth system communication.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

The Smart Campus Event Management System is developed after analyzing several existing event management applications available in the market. These applications provide event organization and registration features but lack certain campus-specific functionalities required by educational institutions.

1. Eventbrite: Eventbrite is a widely used online event management platform that allows users to create, promote, and manage events. It supports ticket booking, attendee management, and email notifications. However, it is mainly designed for commercial events and does not provide institution-specific features such as student login, department-based filtering, or academic event management. Website: <https://www.eventbrite.com>

2. Whova: Whova is an event management and networking platform that offers event registration, agenda scheduling, attendee networking, live announcements, and feedback collection. It is suitable for con-

ferences and seminars but is more focused on large-scale professional events rather than campus event workflows. Website: <https://whova.com>

3. Cvent: Cvent is an enterprise-level event management software that supports event planning, venue selection, registration management, analytics, and email marketing. Although feature-rich, it is costly and complex for small or medium educational institutions. Website: <https://www.cvent.com>

4. RSVPify: RSVPify provides online invitation, RSVP tracking, ticketing, and guest management features. It is simple and user-friendly but lacks academic modules such as student authentication, department categorization, and campus announcement integration. Website: <https://rsvpify.com>

Comparison and Need for Proposed System

Existing systems mainly target business events, conferences, and commercial event management. They do not fully address campus-specific needs such as student registration using institutional credentials, department-wise event categorization, academic notifications, and simplified administrator control. Hence, the Smart Campus Event Management System is proposed as a customized solution specifically designed for educational institutions.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the Smart Campus Event Management System is developed using HTML5, CSS3, and Thymeleaf. HTML5 is used to structure web pages, CSS3 is used for styling and responsive layouts, and Thymeleaf is used as the server-side template engine to dynamically render data from the backend. The frontend is executed through a web browser and integrated with Spring Boot.

3.1.2 Structure of the Project

The frontend consists of multiple pages including login page, registration page, student dashboard, admin dashboard, event listing page, event details page, and feedback forms. Templates are organized under the templates folder and static resources such as CSS files are stored in the static folder. **3.1.2 Structure of the Project**

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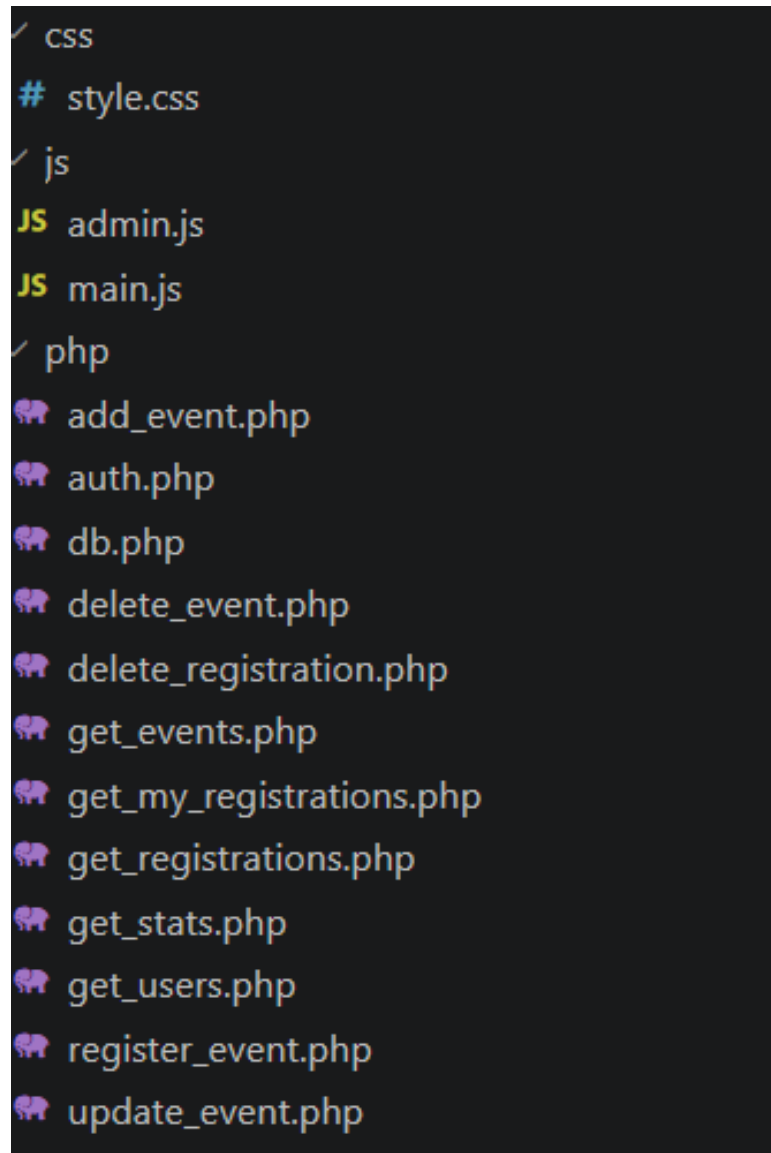


Figure 3.1: Frontend Project Structure Showing Templates, Static Resources, and Web Pages

Figure 3.1 illustrates the frontend project structure of the Smart Campus Event Management System. It contains all frontend components including Thymeleaf templates, CSS files, static resources, and user

interface pages required for application rendering.

3.1.3 Execution Procedure

The frontend is loaded through Spring Boot application startup. Users access the application through browser using localhost. Requests from users are routed to controllers, which process data and return Thymeleaf templates for rendering. add 2 more lines of execution procedure

3.2 Backend Implementation

3.2.1 Environment Setup

The backend is developed using Java with Spring Boot framework. Spring MVC handles request mapping and controller logic, Spring Data JPA manages database operations, and Spring Security provides authentication and authorization. MySQL is used as the database, and Maven is used for dependency management and build automation.

3.2.2 Structure of the Project

The backend follows layered architecture consisting of controller, service, repository, model, dto, config, and exception packages. Controllers handle HTTP requests, services manage business logic, repositories interact with the database, and models represent database entities. This structured architecture improves maintainability, scalability,

and code reusability throughout the application development process.

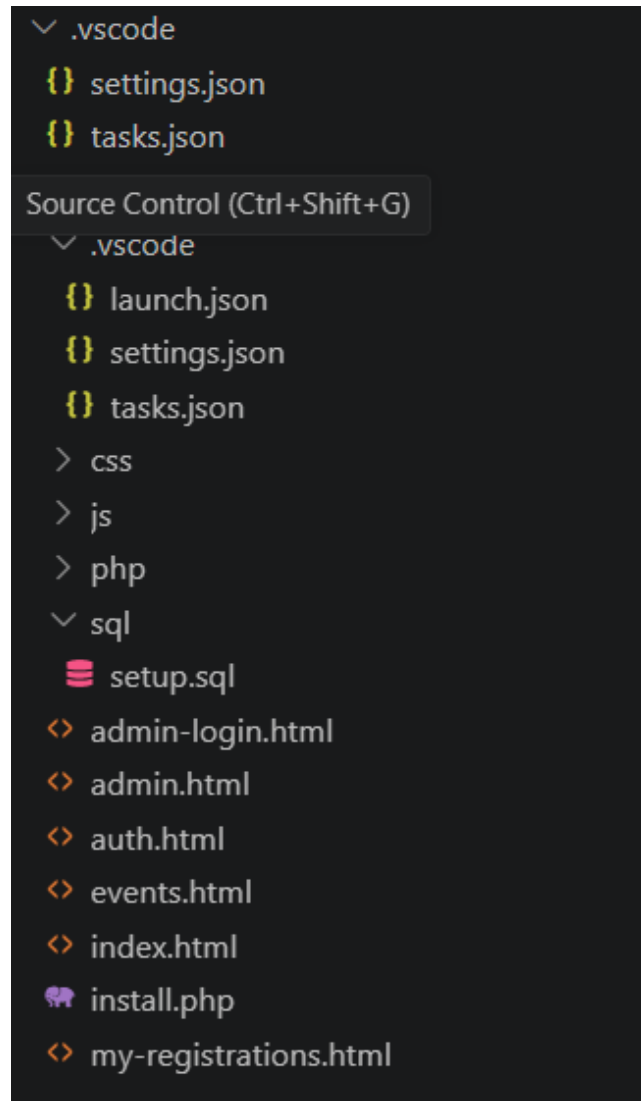


Figure 3.2: Backend Project Structure Showing Controller, Service, and Repository Packages

Figure 3.2 shows the backend project structure of the Smart Campus Event Management System with its main packages and configuration files.

3.2.3 Execution Procedure

The backend execution begins when the Spring Boot application is

started using Maven commands. The embedded Tomcat server initializes the application and loads all required dependencies. User requests are received through controllers, processed by service classes, and database operations are performed using repositories connected to MySQL. Finally, the response is rendered to the frontend using Thymeleaf templates.

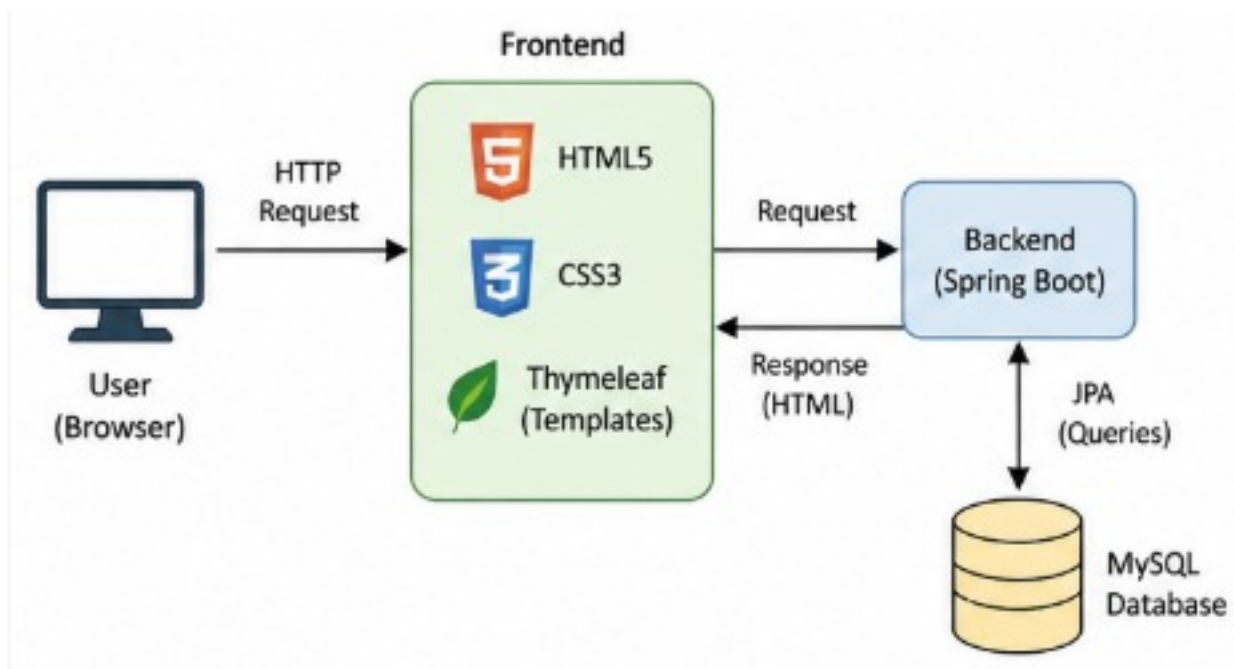


Figure 3.3: Front End Architecture

Figure 3.3 shows the frontend architecture and user interface flow of the system. It ensures smooth navigation and interaction for users.

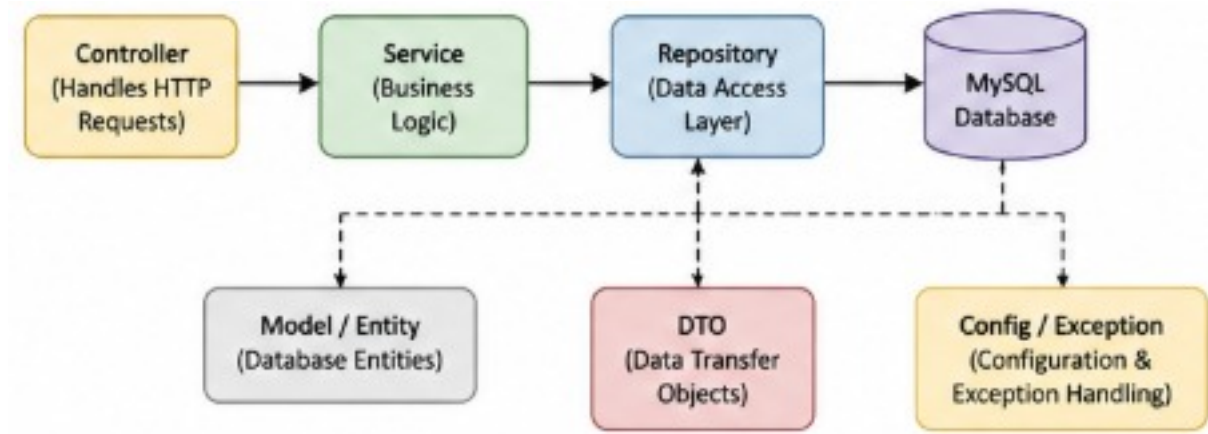


Figure 3.4: Backend Architecture

Figure 3.4 illustrates the backend modules and database connectivity. It manages request processing and business logic execution.

3.3 Database Implementation

3.3.1 Database Configuration

The Smart Campus Event Management System uses MySQL as the relational database management system. Database connectivity is established using Spring Data JPA and JDBC configuration in the application.properties file. The database stores all application data securely, including users, events, registrations, and feedback information.

3.3.2 Schema Structure

The database schema is designed using entity relationships between multiple tables. Core entities include User, Event, Registration, and Feedback. Relationships are maintained using primary keys and for-

eign keys to ensure data consistency and integrity.

Entity	Key Type	Description
User	Primary Key (User_ID)	Uniquely identifies each user and stores login details
Event	Primary Key (Event_ID)	Uniquely identifies each event with details like name and date
Registration	Foreign Key (User_ID, Event_ID)	Links users with events they register for
Feedback	Foreign Key (User_ID, Event_ID)	Stores feedback linked to specific user and event

Table 3.1: Database Schema Structure with Keys and Relationships

Table 3.1 represents the schema design of the database, showing how entities are connected using primary and foreign keys to maintain relational integrity.

3.3.3 Tables created

The main tables created in the database are users, admins, events, registrations, and feedback. These tables store login credentials, event details, registration records, participant information, and student feedback data required for the smooth functioning of the system. The tables are designed to ensure proper data organization and easy retrieval of information. They also help in maintaining data consistency across different modules of the application. Overall, the structured database design supports efficient management of all system operations.

S.No	Table Name	Purpose
1	Users	Stores student/user login details and profile information
2	Admins	Stores administrator credentials and access details
3	Events	Contains event details such as title, date, venue, and description
4	Registrations	Maintains records of student event registrations
5	Feedback	Stores participant feedback and suggestions after events

Table 3.2: Database Tables Used in Smart Campus Event Management System

Table 3.2 presents the major database tables used in the system. These tables are responsible for storing user, event, registration, and feedback details.

Field Name	Sample Data	Description
Username	student123	Unique user login name
Email ID	student@gmail.com	Registered email address
Password	*****	Encrypted user password
Department	CSE	Student department details
Event Name	AI Workshop	Registered event information
Feedback	Excellent Session	Student feedback data

Table 3.3: Sample Data Stored in Database

Table 3.3 shows sample user and event data stored in the database for application functionality.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Smart Campus Event Management System follows a layered MVC architecture consisting of presentation layer, controller layer, service layer, repository layer, and database layer. This architecture ensures modularity, maintainability, and efficient communication between components.

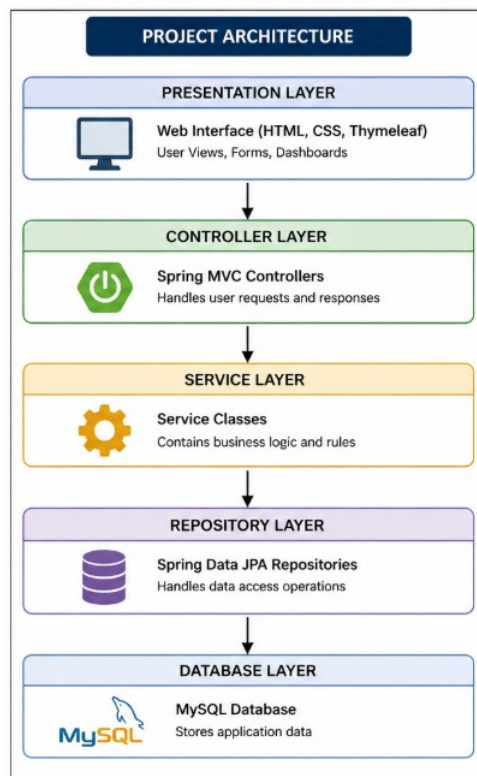


Figure 4.1: Project Architecture

Figure 4.1 illustrates the overall architecture of the Smart Campus Event Management System. It shows the interaction between presentation, business logic, and database layers.

4.2 Data Flow Diagram

The data flow diagram illustrates how user requests are processed through the system. Students and admins interact with the application, requests are processed by the backend, and data is stored or retrieved from the database. The system ensures proper flow of data between frontend, backend, and database components. It clearly defines how inputs are transformed into outputs through different processing stages. This helps in understanding the overall working architecture of the system in a structured manner.

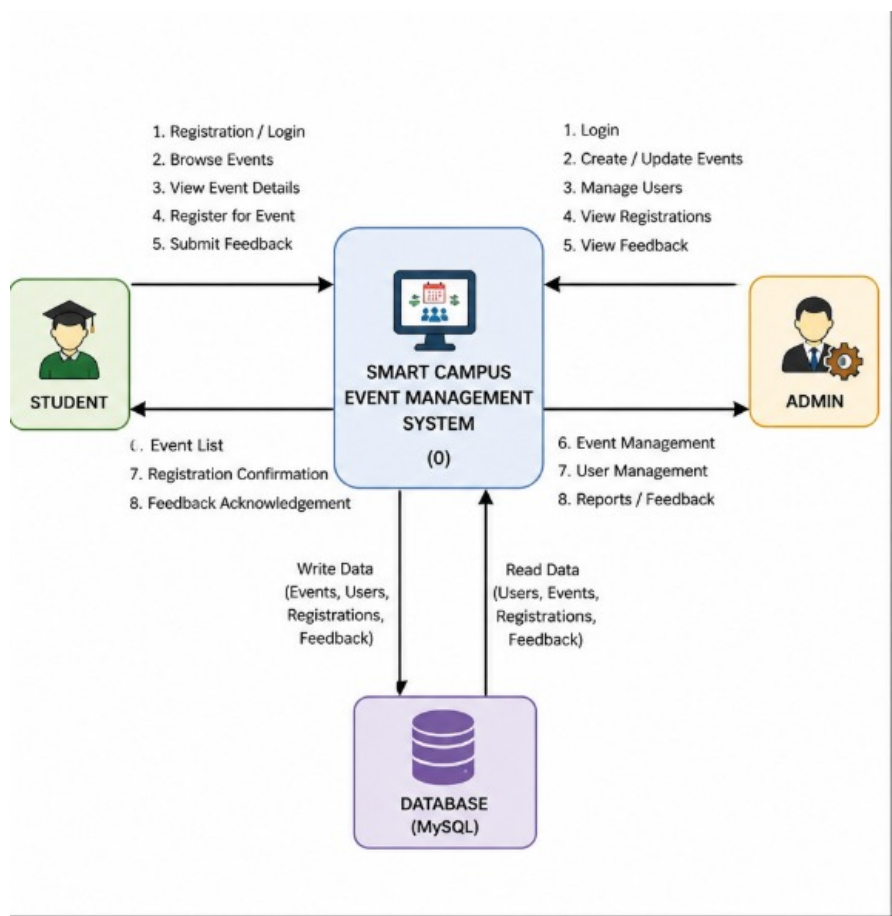


Figure 4.2: Data Flow Diagram

Figure 4.2 represents the flow of data between users, application modules, and database. It explains how requests are processed and responses are generated. This diagram ensures a clear understanding of system workflow.

4.3 Module 1: User Management Module

This module handles the registration and authentication of users such as students, faculty, and administrators. It ensures secure login using credentials and role-based access control. Users can update their profiles, reset passwords, and manage personal information. The module also differentiates permissions based on user roles to maintain system security. It provides a reliable mechanism for validating user access before allowing system operations. This improves data protection and prevents unauthorized access to application resources.

4.4 Module 2: Event Management Module

This module allows administrators and event organizers to create, update, and delete campus events. It includes event details such as title, description, date, time, venue, and capacity. Students can view available events and register for participation. The system also prevents duplicate registrations and manages event schedules efficiently. The module maintains proper event records and participant information. It ensures smooth coordination between organizers and students. Event schedules are managed systematically to avoid conflicts. This module improves overall event planning and execution.

4.5 Module 3: Notification and Feedback Module

This module is responsible for sending real-time notifications and alerts regarding events, reminders, and updates through email or sys-

tem notifications. After event completion, users can provide feedback and ratings. This helps in improving the quality of future events and maintaining engagement between organizers and participants.

The notification system keeps users informed about important announcements. Feedback collection helps organizers analyze participant satisfaction. Ratings and suggestions are stored for future improvements. This module enhances communication and user engagement within the system.

Advantages of this Project

- **Efficient Event Management:** Automates event creation, registration, and scheduling, reducing manual effort.
- **Improved Communication:** Provides instant notifications and alerts to students and faculty.
- **Centralized System:** Maintains all event data in a single platform for easy access and management.
- **Reduced Manual Work:** Minimizes human errors and repetitive administrative tasks.
- **Better Participation:** Encourages students to participate more in campus events.

Disadvantages

- Requires stable internet connection for proper functioning.
- Initial setup and maintenance require technical expertise.
- System performance may reduce under high traffic.
- Users may need initial training to understand system features.

Chapter 5

RESULTS AND DISCUSSIONS

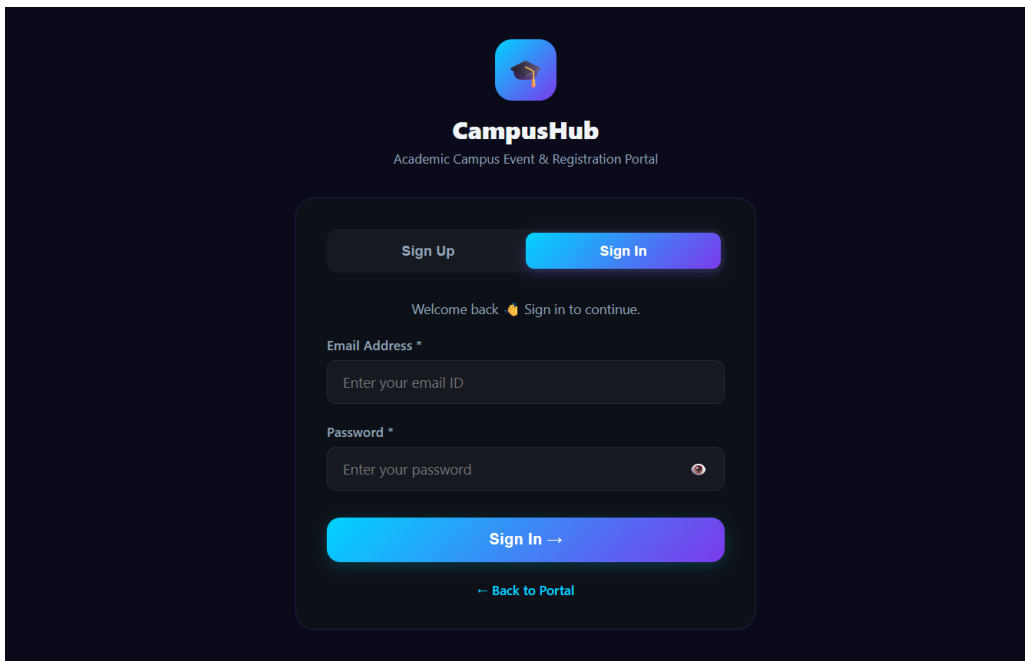


Figure 5.1: Login Page - Admin/Student Authentication

This figure 5.1 shows the login page where both students and administrators enter their credentials to access the system. Based on the role, users are redirected to their respective dashboards. New users must first register by providing details. After successful registration, both students and administrators can sign in using their registered email ID and password to securely access the application. The authentication process validates user credentials before granting access to protected modules. This ensures secure handling of user information and prevents unauthorized access. The login system forms the entry point for accessing all application features.

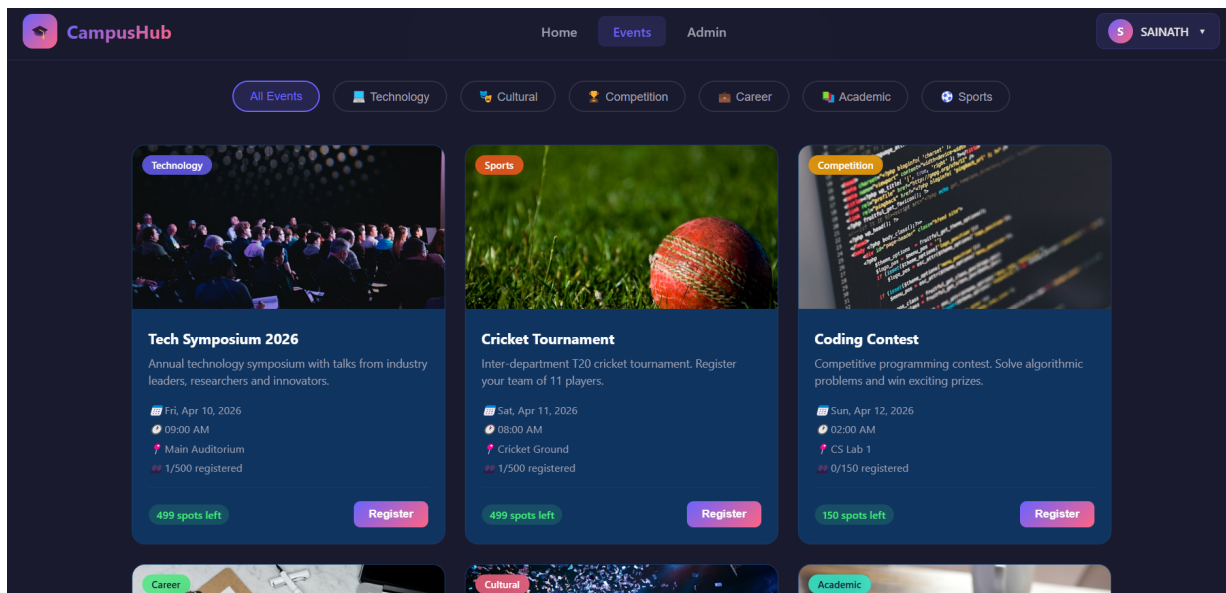


Figure 5.2: Student Dashboard - View and Register Events

This figure 5.2 represents the student dashboard where students can view all available campus events, check event details, and register for participation. The dashboard provides a user-friendly interface for accessing upcoming workshops, seminars, and technical events. Students can also view event deadlines and important announcements related to registrations. The interface is designed to simplify event participation and improve accessibility for students. It helps users stay updated with campus activities through a centralized platform.

- Students can search and browse available events easily.
- Students can register for events using their email ID.
- Students receive a confirmation email after successful registration.

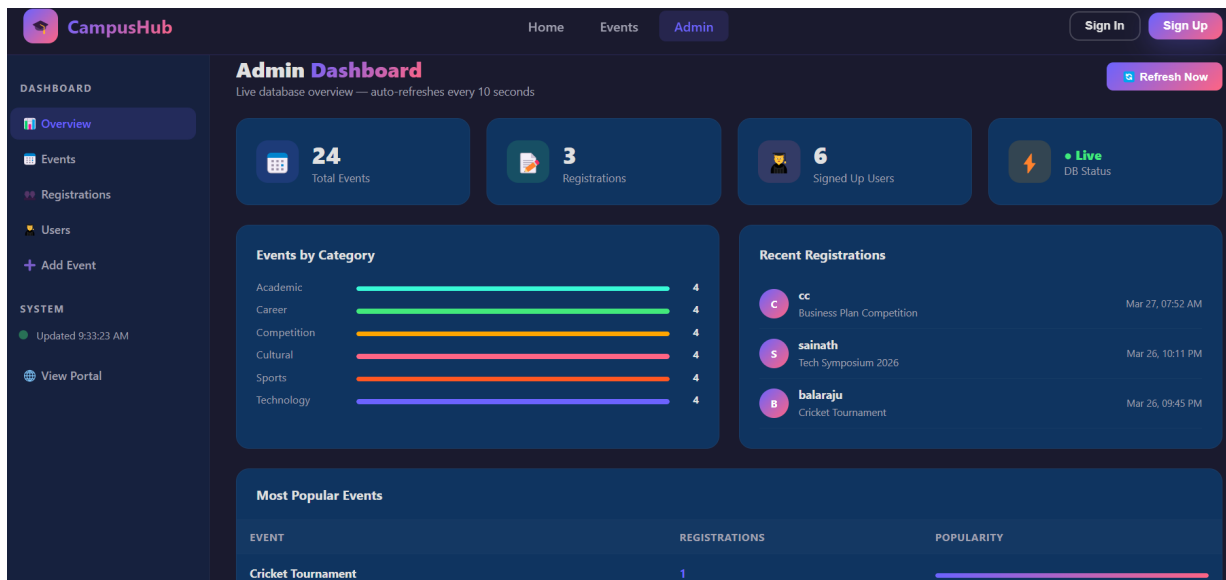


Figure 5.3: Admin Dashboard - Event Management and Registrations

This figure 5.3 shows the admin dashboard where the administrator can create new events, update or delete events, and view student registration details. The dashboard acts as the central management interface for handling all campus event operations. It provides administrators with quick access to event records, participant information, and system controls required for smooth event management.

- Admin can create and manage all campus events including date, time, venue, and description.
- Admin can view and monitor student registrations for each event.
- Admin can track overall event participation and manage registrations efficiently.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Smart Campus Event Management System has been successfully designed and developed to simplify the management of campus events. The system provides an efficient platform for handling user registration, event creation, event participation, notifications, and feedback collection. It reduces manual effort, improves communication between students and administrators, and ensures better organization of events. By centralizing all event-related activities, the system enhances accessibility, accuracy, and overall efficiency in managing campus events. The project also improves transparency in event registrations and enables administrators to monitor activities more effectively. Overall, the system serves as a reliable and scalable solution for digital campus event management. The implementation of secure authentication and role-based access control ensures safe handling of user data. The system architecture supports maintainability and future scalability for additional features. This project demonstrates the practical application of full stack development technologies in solving real-world campus management problems.

6.2 Future Enhancements

In the future, the system can be improved by adding mobile application support for better accessibility. Integration with SMS and WhatsApp notifications can also be implemented for faster communication. Artificial Intelligence-based event recommendations can be added to suggest events based on user interests. Additionally, online payment gateways can be integrated for paid events, and advanced analytics dashboards can be included to analyze student participation and event performance. Cloud deployment can be implemented to improve scalability and system availability. Multi-language support can also be added to make the platform more accessible to users from diverse backgrounds. Integration with college ERP systems can further automate student data synchronization and attendance management. Biometric or QR code-based attendance tracking can also be integrated for event check-ins. Enhanced security mechanisms such as two-factor authentication can be implemented for stronger account protection. Automated report generation can be added to simplify event analysis and administrative decision-making. Social media integration can be introduced for event promotion and wider reach among students. Personalized dashboards can be developed to display recommended events and participation history. Real-time chat support can also be included to improve communication between organizers and participants.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	WKs	Justification
WP1	Depth of knowledge	Requires in depth engineering knowledge	WK3, WK4, WK5, WK6	Requires knowledge in Spring Boot, MVC, database design, REST APIs, and frontend technologies.
WP2	Conflicting requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances usability and security with authentication and validation.
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	Includes system design, schema creation, and validation handling.
WP4	Familiarity	Partially new problems	WK4, WK8	Real-time updates and analytics introduce challenges.
WP5	Applicable Codes	Limited standard solutions	WK6	Custom implementation for filtering and tracking.
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Addresses students, faculty, and administrators.
WP7	Interdependence	System level integration	WK3, WK5, WK6	Integrates UI, backend, database, and security modules.

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	Utilizes Spring Boot, Spring MVC, Spring Data JPA, Thymeleaf, HTML/CSS, and relational databases.
EA2	Level of Interactions	Complex interactions between modules	Handles event management, registration, validation, authentication, and reporting modules.
EA3	Innovation	Application of modern technologies	Web-based centralized platform with filtering and secure admin operations.
EA4	Consequences to Society and Environment	Impact on users and society	Improves engagement and institutional efficiency while reducing errors.
EA5	Familiarity	Beyond standard practices	Combines development, security, and analytics in one system.

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

7.3 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 4	Quality Education	Facilitates participation in academic events, workshops, and seminars.
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation in educational institutions.
SDG 11	Sustainable Cities and Communities	Supports smart campus initiatives and resource management.
SDG 16	Peace, Justice and Strong Institutions	Ensures secure and transparent event handling.
SDG 12	Responsible Consumption and Production	Reduces paper usage through online registration and management.

Table 7.3: **SDG Mapping for the Project**

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